

Predictive models for secondary epilepsy within 1 year in patients with acute ischemic stroke: a multicenter retrospective study

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 Copyright information

Abstract

Objective

Post-stroke epilepsy (PSE) is a significant complication that has a negative impact on the prognosis and quality of life of ischemic stroke patients. We collected medical records from 4 hospitals in Chongqing and created an interpretable machine learning model for prediction.

Methods

We collected medical records, imaging reports, and laboratory tests from 21459 patients with a diagnosis of ischemic stroke. We conducted traditional univariable and multivariable statistics analyses to compare and identify important features. Then the data was divided into a 70% training set and a 30% testing set. We employed the Synthetic Minority Oversampling Technique combined with Edited Nearest Neighbors method to resample an imbalanced dataset in the training set. Nine commonly used methods were used to build machine learning models, and relevant prediction metrics were compared to select the best-performing model. Finally, we used SHAP(SHapley Additive exPlanations) for model interpretability analysis, assessing the contribution and clinical significance of different features to the prediction.

Results

In the traditional regression analysis, complications such as hydrocephalus, cerebral hernia, uremia, deep vein thrombosis; significant brain regions included the involvement of the cortical regions including frontal lobe, parietal lobe, occipital lobe, temporal lobe, subcortical region of basal ganglia, thalamus and so on contributed to PSE. General features such as age, gender, and the National Institutes of Health Stroke Scale score, as well as laboratory indicators including WBC count, D-dimer, lactate, HbA1c and so on were associated with a higher likelihood of PSE. Patients with conditions such as fatty liver, coronary heart disease, hyperlipidemia, and low HDL had a higher likelihood of developing PSE. The machine learning models, particularly tree models such as Random Forest, XGBoost, and LightGBM, demonstrated good predictive performance with an AUC of 0.99.

Conclusion

The model built on a large dataset can effectively predict the likelihood of PSE, with tree-based models performing the best. The NIHSS score, WBC count and D-dimer were found to have the greatest impact.

eLife Assessment

This **valuable** study reports machine learning models derived from large-scale data to predict the risk of post-stroke epilepsy. The evidence supporting the conclusions is **convincing**, although there are some validation issues (lack of cross-validation, possible bias in external validation results). The study may be of interest in the field of clinical neurology

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Introduction

Stroke is the second leading cause of death worldwide, with an annual mortality rate of approximately 5.5 million, and also the leading cause of disability globally, accounting for 50% of cases [1]. Generally, ischemic stroke accounts for the majority, about 80% of stroke cases [2] [3]. Post-stroke epilepsy (PSE) is a significant complication, with studies indicating that as many as 3-30% of stroke patients develop epilepsy, which has a negative impact on patients' prognosis and quality of life [4]. It can exacerbate cognitive, psychiatric, and physical impairments caused by cerebrovascular disease and comorbidities [5]. Furthermore, the highest incidence of PSE occurs within the first year after acute stroke, accounting for about half of the cases [2]. Therefore, early prediction and intervention for PSE, especially ischemic ones, are crucial.

Currently, most studies utilize clinical data to establish statistical models, survival analysis and cox regression [2] [6], and multiple linear regression [7] to construct simple models for the prediction of PSE. Last year, Lin et al. developed a model based on radiomics that outperformed the conventional clinical model in predicting PSE related to intracerebral hemorrhage (ICH). They suggested that a combined radiomics-clinical model could better assist clinicians in assessing the

individual risk of PSE after the first occurrence of ICH and facilitate early diagnosis and treatment of PSE [8]. However, subsequent studies have raised doubts regarding the application of radiomics, suggesting the need for further research [9]. Overall, there is still a relative scarcity of research on PSE prediction, with most studies focusing on the analysis of specific or certain risk factors [10][11][8][12] constructing simple models and hardly proposed or established a more comprehensive and scientifically accurate prediction model.

Machine learning has emerged as a promising approach in recent years for constructing medical models, as it excels in handling large volumes of data and complex information, and has been increasingly applied in neuroscience and clinical prediction [13][14][15]. Previous studies have utilized machine learning for related research on post-stroke cognitive impairments [16], stroke and myocardial infarction risk prediction models in large artery vasculitis patients [14], post-stroke depression prediction models based on liver function test indicators [17], and prediction of hematoma expansion in traumatic brain injury (TBI) [18]. Models constructed using machine learning algorithms can automatically handle linear or complex nonlinear relationships between different variables and provide insights into the contribution of different features to the prediction target, which is challenging for traditional statistical models. However, machine learning methods require a substantial amount of data and are prone to overfitting when trained on small sample data. The more valid and high-quality data input, the better machine learning algorithms can capture the underlying patterns between the data, thereby achieving more accurate predictions.

This study try to select important risk factors from mutiple fearures extracted from the clinical records and examination data of ischemic stroke patients and subsequently develops a prediction model for PSE using machine learning methods. By utilizing relevant early admission features of ischemic stroke patients, we aim to automatically predict the probability of PSE occurrence and further guide clinical decision-making and nursing care.

Research content and method

Research patients

This study retrospectively included all stroke patients admitted to the Chongqing Emergency Center between June 2017 and June 2022 for the development of the prediction model. Subsequently, patient data from three external validation centers, namely, Qianjiang Central Hospital, Bishan District People's Hospital, and Yubei District Traditional Chinese Medicine Hospital, were collected between July 2022 and July 2023 for external validation and evaluation of the model. The external validation cohort focused more on collecting positive cases to examine the model's ability to identify positive samples.

Inclusion criteria: (1) Age between 18 and 90 years at admission; (2) Diagnosed with acute ischemic stroke and hospitalized for treatment.

Exclusion criteria: (1) Patients with a history of stroke or transient ischemic attack (TIA); (2) Patients with a history of other conditions such as traumatic brain injury, intracranial tumors, or cerebral vascular malformations that may cause epilepsy; (3) Patients with a history of epilepsy or who have received antiseizure medications for the prevention of seizures or for other diseases (such as migraine or psychiatric disorders); (4) Patients who died within 72 hours after stroke onset.

This study collected de-identified data from relevant patients for the construction of a multi-modal database for stroke patients. The study protocol was approved by the Ethics Committees of Chongqing University Center Hospital, Chongqing University Qianjiang Central Hospital, Bishan

District People's Hospital, and Yubei Traditional Chinese Medicine Hospital.

The procedure of selection is in [figure1](#). Total there are 42079 records from the stroke database, 24733 patients were diagnosed as ischemic stroke or lacunar stroke with new onset. Then we excluded hemorrhage stroke(4565), history of stroke(2154), TIA(3570), unclear cause stroke(561) and records who missed important data(6496). Then we excluded patients whose seizure might be attributed to other potential causes (brain tumor, intracranial vascular malformation, traumatic brain injury, etc)(865). Then we excluded patient who had a seizure history(152) or died in hospital (1444). Then we excluded patients who were lost to follow-up (had no outpatient records and can't contact by phone) or died within 3 months of the stroke incident(813). Finally 21459 cases are involved in this research.

Data collection

We extracted all records and other relevant data from the database of the hospitals. Under the structure of PostgreSQL we coded Structured Query Language to manage different data as follows:

1. General information: gender, age, NIHSS(the National Institutes of Health Stroke Scale) score at admission;
2. (2) Comorbidities and complications: uremia, DVT(previous deep vein thrombosis), diabetes mellitus, hypertension, coronary atherosclerosis, atrial fibrillation, cerebral hernia, hydrocephalus, hypoproteinemia, hyperuricemia, hyperlipidemia, internal carotid stenosis, common carotid stenosis, etc.
3. According to CT or MRI records, the patient's cortical lobes and subcortical involvement were counted: frontal lobe \ parietal lobe \ temporal lobe \ occipital lobe \ insular lobe \ basal ganglia \ internal capsule \ brain stem \ cerebellum \ periventricular \ centrum semiovale \ thalamus involvement. In addition, the extent of cortical involvement (frontal lobe, parietal lobe, temporal lobe, occipital lobe and insular lobe each accumulated 1 point) and the extent of subcortical involvement (basal ganglia, internal capsule, brain stem, periventricular, thalamus and cerebellum any accumulated 1 point) were summarized.
4. According to CTA, MRA or DSA records, the patient's vascular stenosis or occlusion was counted: ACA(anterior cerebral artery) \ MCA(middle cerebral artery) \ PCA(posterior cerebral artery) \ VA(vertebral artery) \ BA(basilar artery)
5. Important laboratory indicators: Blood lipids (TG(Triglyceride), HDL(High Density Lipoprotein Cholesterol), LDL(Low Density Lipoprotein Cholesterol)), liver function (ALT(Alanine Transaminase), AST(Aspartate Aminotransferase), Bilirubin, Albumin), renal function (Urea, BUA(Blood Uric Acid), Creatinine), blood gas (Lactate, Anion Gap, TCO₂(Total Carbon Dioxide)), coagulation related indicators (INR(International Normalized Ratio), PT(Prothrombin Time), APTT(Activated Partial Thromboplastin Time), TT(Thrombin Time), D-Dimer, Fibrinogen) and myocardial enzymes (CK(Creatine Kinase), CK-MB(Creatine Kinase Isoenzyme), LDH(Lactate Dehydrogenase), IMA(Ischemic Modified Albumin), HBDH(α -Hydroxybutyrate Dehydrogenase)).

Data processing and model building

(Processing of missing data) We counted the values of all laboratory indicators for the first time after stroke admission (everyone who was admitted because of stroke would perform blood routine, liver and kidney function and so on), excluded indicators with missing values of more than 10%, and filled the data of the remaining indicators with missing values by random forest algorithm using the default parameter. First, we go through all the features, starting with the one with the least missing (since the least accurate information is needed to fill in the feature with the

Figure 1.

Selection and exclusion procedure of patients

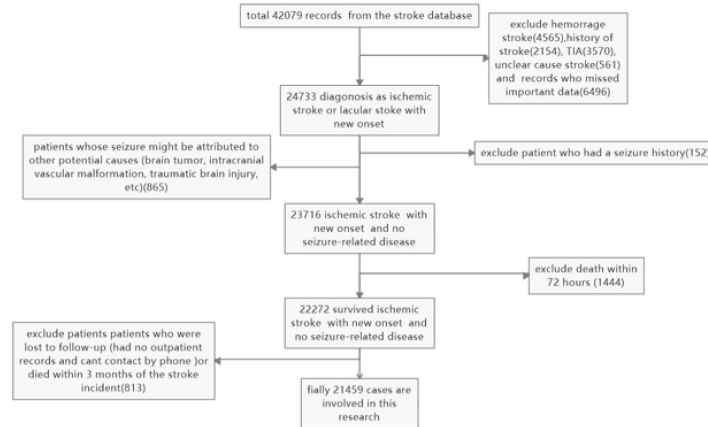
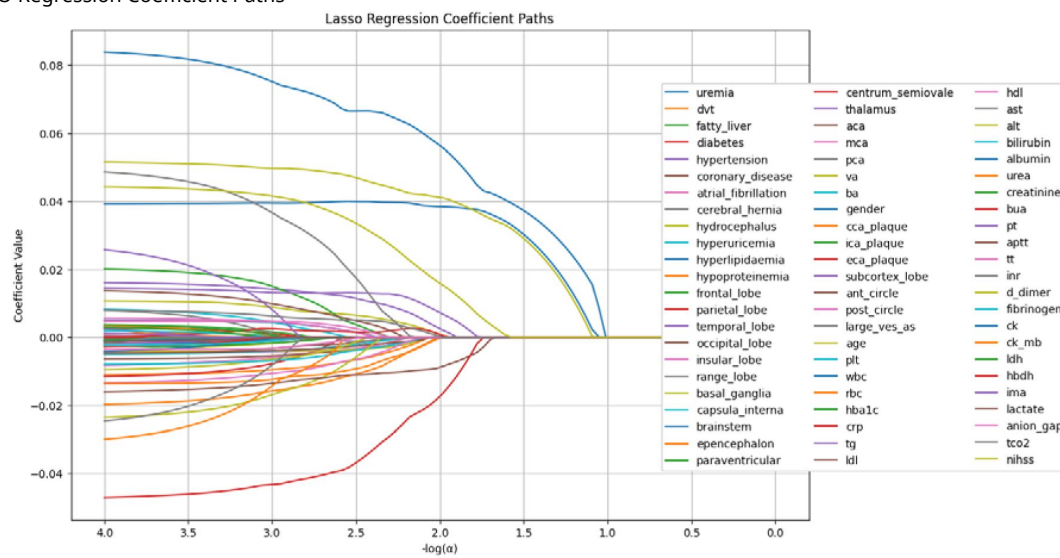


Figure 2.

LASSO Regression Coefficient Paths



least missing). When filling in a feature, replace the missing value of the other feature with 0. Each time a regression prediction is completed, the predicted value is placed in the original feature matrix and the next feature is filled in. After going through all the features, the data is complete.

(Distribution of characteristics) Univariate analysis was used to examine the distribution of characteristics between the PSE negative group and the positive group. The data were then divided into a training set and a test set at a ratio of 7:3.

(Processing of unbalanced data) Considering the low incidence of PSE and the small proportion of positive patients, the positive data of the training set were augmented by Synthetic Minority Over-sampling Technique combined with Edited Nearest Neighbors by using default parameter of SMOTEENN method from imblearn python package and set random seed at 42 for repetition.

(Processing of categorical data) For categorical data, the one-hot method is used for transformation. The LASSO method was then used in the training set to screen the important features.

(Model building) We first used LASSO regression to select the 20 most important features. Next, we employed 9 common machine learning methods, including Naive Bayes, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Multi-Layer Perceptron, XGBoost, LightGBM, and K-Nearest Neighbors. We then optimized the hyperparameters of each model through grid search to improve their performance. To evaluate the models, we calculated metrics such as accuracy, sensitivity, specificity, F1-score, positive predictive value, and negative predictive value. We also plotted the ROC curve, calibration curve, and decision curve. Additionally, we used an independent external validation dataset to assess the generalization performance of the selected model. Finally, we leveraged the SHAP algorithm to perform an interpretable analysis of the best-performing model, investigating the contribution of each feature to the model's predictions and their clinical significance. Through this series of model building, optimization, and analysis, we developed a machine learning model with good predictive performance and interpretability, providing valuable support for clinical decision-making.

Statistical approach

PostgreSQL v15 (<http://www.postgresql.org/>) was used to search and extract the data from the local database.

The open-source statistical package “Scipy.stats” in Python was used for statistical analysis. The details of the univariate significance analysis for each feature are as follows:

The Shapiro-Wilk test was used to check the normality of the distribution for each feature. For features that did not follow a normal distribution, the Mann-Whitney U test was used to assess their significance with respect to the target variable.

For features that exhibited a normal distribution, the Levene test was employed to assess the homogeneity of variances. Features with homogeneous variances were analyzed using the Student's t-test to determine their significance with respect to the target variable, while features with heterogeneous variances were analyzed using the Welch's t-test.

The confidence intervals for the AUC values and Brier scores were obtained by performing 1000 bootstrap resampling iterations on the corresponding datasets. The binary classification thresholds for the predicted probabilities generated by all models were established using the maximum Youden index derived from the training cohort.

Throughout the study, a two-tailed p-value less than 0.05 was considered statistically significant.

All the codes were uploaded at <https://github.com/conanan/lasso-ml>.

Result

Filling of missing data

These features had missing values that were filled using a Random Forest (RF) model, addressing the missing data one feature at a time: Plt, WBC, RBC, HbA1c, CRP, TG, LDL, HDL, AST, ALT, Bilirubin, Albumin, Urea, Creatinine, BUA, PT, APTT, TT, INR, D-dimer, Fibrinogen, CK, CK-MB, LDH, HBDH, IMA, Lactate, Anion_gap, TCO2, NIHSS.

Characteristics of study participants

A total of 21459 patients were included in this study, of which 15021 patients were included in the training set, and the incidence of PSE was 4.3%. The test set contained 6438 patients with a PSE incidence of 4.3%. The external validation cohort consisted of 536 patients at three hospitals. Statistical details of the clinical characteristics of the patients are provided in the [table1](#).

Statistical analysis showed that the patients who had higher possibility of PSE were with complications of uremia, history of DVT, atrial fibrillation, hyperuricemia, cerebral hernia and hydrocephalus. The involved locations of frontal lobe, parietal lobe, occipital lobe, temporal lobe, cortex, subcortex, basal ganglia and hypothalamus. The general characteristics included age, gender, nihss score; Laboratory indicators included WBC count, HbA1C, CRP, triglycerides, AST, ALT, bilirubin, urea, uric acid, APTT, PT, D-dimer, CK, CK-MB, LDH, HBDH, IMA, lactate, and anion gap. Besides, the p values of fatty liver, coronary heart disease, hyperlipidemia, and hdl were significant, and patients with negative or low values of these indicators had a high risk of secondary disease. The statistics analysis result, the uni and multi regression analysis result table is in [table1](#), [table2](#) and [table3](#).

Performance of machine learning models

The relevant indicators of the machine learning model are shown in [table4](#), and the ROC curves, calibration curve and DCA are shown in [figure3](#). It can be found that the over all models the AUC of tree models such as RF, XGboost and lightGBM are better than other models, and the PPV value of random forest is the highest, reaching 0.864, which is the most important function of our models. Complex machine learning algorithms were superior to traditional logistic regression. The Brier score of the calibration curve reached 0.006, and the DCA also showed good clinical decision-making benefits, which had good practical value. In the external validation cohort, we use the RF to predict. The Sensitivity was 0.91, the PPV was 0.95, demonstrate a good predictive ability of the model.

Analysis of SHAP risk factors

The analysis in [Figure 4](#) shows the SHAP (Shapley Additive Explanations) values, individual decision attempts, and overall decision curves. Among the general characteristics, females had a higher rate of PSE. Regarding the NIHSS score, higher score cause higher incidence rate of PSE. Higher values of WBC count, D-dimer, CRP, AST, CK-MB, HbA1c, bilirubin, TCO2, and LDH at admission were associated with a greater likelihood of developing PSE. Conversely, lower values of HBDH, PLT, and APTT were also linked to a higher probability of the outcome. However, the specific regions of the brain affected did not have a significant individual effect on the overall outcome. Among the complications, only hypertension was more strongly associated with the development of the outcome. Other conditions, such as coronary heart disease, diabetes, hyperlipidemia, and fatty liver, were less likely to be related to the outcome. We use the force plot

Feature	positive, N=954	negative, N = 20789	method	P	stats
eca_plaque	-	-	Chi-Square	0.438971	0.59897
—0	942 (98.742%)	20591 (99.048%)	-	-	-
—1	12 (1.258%)	198 (0.952%)	-	-	-
subcortex_lobe	-	-	Chi-Square	0.001273	10.381551
—0	814 (85.325%)	18454 (88.768%)	-	-	-
—1	140 (14.675%)	2335 (11.232%)	-	-	-
ba	-	-	Chi-Square	0.991017	0.000127
—0	945 (99.057%)	20605 (99.115%)	-	-	-
—1	9 (0.943%)	184 (0.885%)	-	-	-
hypertension	-	-	Chi-Square	0.602539	0.271184

Table 1.

Single factor significant analysis results

	—0	290 (30.398%)	6497 (31.252%)	-	-	-
	—1	664 (69.602%)	14292 (68.748%)	-	-	-
ica_plaque		-	-	Chi-Square	0.152086	2.051203
	—0	878 (92.034%)	19392 (93.28%)	-	-	-
	—1	76 (7.966%)	1397 (6.72%)	-	-	-
frontal_lobe		-	-	Chi-Square	0	53.171781
	—0	868 (90.985%)	19943 (95.931%)	-	-	-
	—1	86 (9.015%)	846 (4.069%)	-	-	-
cerebral_hernia		-	-	Chi-Square	0.000032	17.284355
	—0	934 (97.904%)	20626 (99.216%)	-	-	-
	—1	20 (2.096%)	163 (0.784%)	-	-	-
thalamus		-	-	Chi-Square	0.060918	3.512207
	—0	937 (98.218%)	20565 (98.923%)	-	-	-
	—1	17 (1.782%)	224 (1.077%)	-	-	-
occipital_lobe		-	-	Chi-Square	0.000034	17.17679
	—0	919 (96.331%)	20422 (98.235%)	-	-	-
	—1	35 (3.669%)	367 (1.765%)	-	-	-
pca		-	-	Chi-Square	0.891182	0.018717
	—0	952 (99.79%)	20729 (99.711%)	-	-	-
	—1	2 (0.21%)	60 (0.289%)	-	-	-
paraventricular		-	-	Chi-Square	0.213759	1.545786
	—0	899 (94.235%)	19786 (95.175%)	-	-	-
	—1	55 (5.765%)	1003 (4.825%)	-	-	-
mca		-	-	Chi-Square	0.393066	0.729435
	—0	912 (95.597%)	19998 (96.195%)	-	-	-
	—1	42 (4.403%)	791 (3.805%)	-	-	-
coronary_disease		-	-	Chi-Square	0	26.19087

Table 1. (continued)

	—0	599 (62.788%)	11288 (54.298%)	-	-	-
	—1	355 (37.212%)	9501 (45.702%)	-	-	-
hypoproteinemia		-	-	Chi-Square	0	53.351931
	—0	774 (81.132%)	18479 (88.888%)	-	-	-
	—1	180 (18.868%)	2310 (11.112%)	-	-	-
parietal_lobe		-	-	Chi-Square	0	57.137771
	—0	884 (92.662%)	20180 (97.071%)	-	-	-
	—1	70 (7.338%)	609 (2.929%)	-	-	-
aca		-	-	Chi-Square	0.928981	0.007944
	—0	941 (98.637%)	20524 (98.725%)	-	-	-
	—1	13 (1.363%)	265 (1.275%)	-	-	-
brainstem		-	-	Chi-Square	0.294979	1.096759
	—0	938 (98.323%)	20532 (98.764%)	-	-	-
	—1	16 (1.677%)	257 (1.236%)	-	-	-
hyperuricemia		-	-	Chi-Square	0.000001	25.147468
	—0	801 (83.962%)	18547 (89.215%)	-	-	-
	—1	153 (16.038%)	2242 (10.785%)	-	-	-
temporal_lobe		-	-	Chi-Square	0	57.872112
	—0	886 (92.872%)	20209 (97.21%)	-	-	-
	—1	68 (7.128%)	580 (2.79%)	-	-	-
diabetes		-	-	Chi-Square	0.389926	0.739172
	—0	617 (64.675%)	13737 (66.078%)	-	-	-
	—1	337 (35.325%)	7052 (33.922%)	-	-	-
range_lobe		-	-	Chi-Square	0	85.377485
	—0	830 (87.002%)	19559 (94.083%)	-	-	-
	—1	43 (4.507%)	467 (2.246%)	-	-	-
	—2	32 (3.354%)	329 (1.583%)	-	-	-

Table 1. (continued)

—3	31 (3.249%)	224 (1.077%)	-	-	-
—4	15 (1.572%)	175 (0.842%)	-	-	-
—5	3 (0.314%)	35 (0.168%)	-	-	-
epencephalon	-	-	Chi-Square	1	0
—0	934 (97.904%)	20362 (97.946%)	-	-	-
—1	20 (2.096%)	427 (2.054%)	-	-	-
hydrocephalus	-	-	Chi-Square	0	181.23517
—0	895 (93.816%)	20565 (98.923%)	-	-	-
—1	59 (6.184%)	224 (1.077%)	-	-	-
insular_lobe	-	-	Chi-Square	0.391042	0.735699
—0	938 (98.323%)	20519 (98.701%)	-	-	-
—1	16 (1.677%)	270 (1.299%)	-	-	-
gender	-	-	Chi-Square	0	44.244052
—0	372 (38.994%)	10407 (50.06%)	-	-	-
—1	582 (61.006%)	10382 (49.94%)	-	-	-
uremia	-	-	Chi-Square	0.00008	15.568169
—0	934 (97.904%)	20618 (99.177%)	-	-	-
—1	20 (2.096%)	171 (0.823%)	-	-	-
atrial_fibrillation	-	-	Chi-Square	0.008017	7.029734
—0	838 (87.841%)	18811 (90.485%)	-	-	-
—1	116 (12.159%)	1978 (9.515%)	-	-	-
centrum_semiovale	-	-	Chi-Square	0.36206	0.830735
—0	922 (96.646%)	20207 (97.2%)	-	-	-
—1	32 (3.354%)	582 (2.8%)	-	-	-
basal_ganglia	-	-	Chi-Square	0.005355	7.755329
—0	893 (93.606%)	19869 (95.575%)	-	-	-
—1	61 (6.394%)	920 (4.425%)	-	-	-

Table 1. (continued)

dvt	-	-	Chi-Square	0	40.790867
—0	847 (88.784%)	19534 (93.963%)	-	-	-
—1	107 (11.216%)	1255 (6.037%)	-	-	-
fatty_liver	-	-	Chi-Square	0.000171	14.123893
—0	812 (85.115%)	16655 (80.114%)	-	-	-
—1	142 (14.885%)	4134 (19.886%)	-	-	-
hyperlipidaemia	-	-	Chi-Square	0.000317	12.969155
—0	801 (83.962%)	16439 (79.075%)	-	-	-
—1	153 (16.038%)	4350 (20.925%)	-	-	-
cca_plaque	-	-	Chi-Square	0.376965	0.780577
—0	751 (78.721%)	16100 (77.445%)	-	-	-
—1	203 (21.279%)	4689 (22.555%)	-	-	-
va	-	-	Chi-Square	0.797483	0.065847
—0	927 (97.17%)	20159 (96.97%)	-	-	-
—1	27 (2.83%)	630 (3.03%)	-	-	-
fibrinogen	3.518 ± 0.663	3.602 ± 0.464	Mann-Whitney U	0.434584	10064078.5
d_dimer	4.362 ± 4.398	1.198 ± 0.98	Mann-Whitney U	0	3555180.5
bua	342.521 ± 74.651	344.132 ± 58.336	Mann-Whitney U	0.000037	10698805.5
tco2	22.739 ± 1.025	22.781 ± 1.225	Mann-Whitney U	0.166751	10178363
hbdh	209.295 ± 57.826	175.906 ± 48.18	Mann-Whitney U	0	6107843
anion_gap	13.026 ± 1.456	12.345 ± 1.368	Mann-Whitney U	0	6496800
ldl	2.686 ± 0.372	2.685 ± 0.361	Mann-Whitney U	0.23394	10140916.5

Table 1. (continued)

tt	16.636 ± 0.809	16.432 ± 0.615	Mann-Whitney U	0	7950954.5
nihss	11.529 ± 2.564	7.886 ± 2.871	Mann-Whitney U	0	2984725.5
albumin	40.734 ± 2.37	40.886 ± 2.257	Mann-Whitney U	0.025821	10338834.5
inr	1.068 ± 0.072	1.076 ± 0.149	Mann-Whitney U	0	9016933.5
tg	1.662 ± 0.484	1.536 ± 0.433	Mann-Whitney U	0	7582690.5
bilirubin	16.516 ± 4.009	15.197 ± 3.981	Mann-Whitney U	0	7522775
ima	81.624 ± 8.559	75.458 ± 12.891	Mann-Whitney U	0	4487861
pt	13.822 ± 0.627	13.843 ± 1.151	Mann-Whitney U	0	8374380.5
crp	55.681 ± 48.823	15.314 ± 18.865	Mann-Whitney U	0	3060302
wbc	11.79 ± 3.084	8.316 ± 1.286	Mann-Whitney U	0	2667973
age	65.335 ± 13.909	66.806 ± 12.597	Mann-Whitney U	0.013188	10386092
hdl	1.246 ± 0.146	1.249 ± 0.149	Mann-Whitney U	0.619502	10008026
lactate	2.825 ± 0.376	2.505 ± 0.411	Mann-Whitney U	0	4480425
rbc	4.408 ± 0.274	4.304 ± 0.324	Mann-Whitney U	0	7811417
ast	38.25 ± 18.205	26.05 ± 12.823	Mann-Whitney U	0	3814876
plt	180.251 ± 36.939	190.132 ± 26.424	Mann-Whitney U	0	11826502.5
alt	26.827 ± 10.349	24.193 ± 10.108	Mann-Whitney U	0	7632233.5

Table 1. (continued)

aptt	35.045 ± 1.881	35.702 ± 2.313	Mann-Whitney U	0	11737054.5
ldh	296.455 ± 111.282	215.357 ± 75.036	Mann-Whitney U	0	5261997.5
creatinine	83.837 ± 24.574	85.199 ± 52.439	Mann-Whitney U	0	8567930.5
hba1c	6.759 ± 1.048	6.662 ± 0.916	Mann-Whitney U	0.000035	9132523
urea	6.33 ± 1.354	6.419 ± 1.438	Mann-Whitney U	0.001566	10515532
ck	1029.594 ± 872.8	195.007 ± 273.212	Mann-Whitney U	0	3469376

Table 1. (continued)

Feature	0 (N=207 89)	1 (N=954)	OR (univariabl e)	coe f	std err	z	P > z	[0. 02 5	0.9 75]	Label_1	Label_0
age	66.806 ± 12.597	65.335 ± 13.909	0.991 (0.986- 0.996, p=0.0)	- 0.0 090	- 0.0 03	- 3.5 08	0. 00 0	- 0.0 14	- 0.0 04	-	-
plt	190.13 2 ± 26.424	180.25 1 ± 36.939	0.986 (0.983- 0.988, p=0.0)	- 0.0 141	- 0.0 01	- 11. 32 0	0. 00 0	- 0.0 17	- 0.0 12	-	-
wbc	8.316 ± 1.286	11.79 ± 3.084	2.23 (2.149- 2.314, p=0.0)	0.8 022	0.0 19	42. 30 6	0. 00 0	0.7 65	0.8 39	-	-
rbc	4.304 ± 0.324	4.408 ± 0.274	2.622 (2.162- 3.177, p=0.0)	0.9 638	0.0 98	9.8 05	0. 00 0	0.7 71	1.1 56	-	-
hba1c	6.662 ± 0.916	6.759 ± 1.048	1.112 (1.042- 1.186, p=0.001)	0.1 059	0.0 33	3.1 76	0. 00 1	0.0 41	0.1 71	-	-
crp	15.314 ± 18.865	55.681 ± 48.823	1.033 (1.031- 1.035,	0.0 326	0.0 01	36. 79	0. 00	0.0 31	0.0 34	-	-

Table 2.

Univariable logistic regression results

			p=0.0)			2	0					
			1.617 (1.441-				0.					
tg	1.536 ± 0.433	1.662 ± 0.484	1.815, p=0.0)	0.4 807	0.0 59	8.1 70	00 0	0.3 65	0.5 96	-	-	
			1.009 (0.843-				0.	-				
ldl	2.685 ± 0.361	2.686 ± 0.372	1.207, p=0.924)	0.0 087	0.0 91	0.0 95	92 4	0.1 71	0.1 88	-	-	
			0.87 (0.562-	-		-	0.	-				
hdl	1.249 ± 0.149	1.246 ± 0.146	1.349, p=0.534)	0.1 389	0.2 23	0.6 22	53 4	0.5 77	0.2 99	-	-	
			1.028 (1.024-			17.	0.					
ast	26.05 ± 12.823	38.25 ± 18.205	1.031, p=0.0)	0.0 277	0.0 02	00 7	00 0	0.0 24	0.0 31	-	-	
			1.017 (1.012-				0.					
alt	24.193 ± 10.108	26.827 ± 10.349	1.021, p=0.0)	0.0 169	0.0 02	7.5 07	00 0	0.0 12	0.0 21	-	-	
			1.068 (1.054-				0.					
bilirubin	15.197 ± 3.981	16.516 ± 4.009	1.082, p=0.0)	0.0 662	0.0 07	9.8 26	00 0	0.0 53	0.0 79	-	-	
			0.971 (0.945-	-		-	0.	-	-			
albumin	40.886 ± 2.257	40.734 ± 2.37	0.999, p=0.042)	0.0 291	0.0 14	2.0 36	04 2	0.0 57	0.0 01	-	-	
			0.955 (0.91-	-		-	0.	-				
urea	6.419 ± 1.438	6.33 ± 1.354	1.002, p=0.063)	0.0 459	0.0 25	1.8 62	06 3	0.0 94	0.0 02	-	-	
			0.999 (0.998-	-		-	0.	-				
creatinine	85.199 ± 52.439	83.837 ± 24.574	1.001, p=0.425)	0.0 006	0.0 01	0.7 98	42 5	0.0 02	0.0 01	-	-	
			1.0 (0.998-	-		-	0.	-				
bua	344.13 2 ± 58.336	342.52 1 ± 74.651	1.001, p=0.411)	0.0 005	0.0 01	0.8 22	41 1	0.0 02	0.0 01	-	-	
			0.982 (0.925-	-		-	0.	-				
pt	13.843 ± 1.151	13.822 ± 0.627	1.043, p=0.564)	0.0 177	0.0 31	0.5 77	56 4	0.0 78	0.0 42	-	-	

Table 2. (continued)

aptt	35.702 ± 2.313	35.045 ± 1.881	0.863 (0.835- 0.891, p=0.0)	- 0.1 473	0.0 17	- 8.9 17	0. 00 0	- 0.1 80	- 0.1 15	-	-
tt	16.432 ± 0.615	16.636 ± 0.809	1.411 (1.287- 1.547, p=0.0)	0.3 442	0.0 47	7.3 28	0. 00 0	0.2 52	0.4 36	-	-
inr	1.076 ± 0.149	1.068 ± 0.072	0.643 (0.385- 1.074, p=0.091)	- 0.4 421	0.2 62	- 1.6 89	0. 09 1	- 0.9 55	- 0.0 71	-	-
d_dimer	1.198 ± 0.98	4.362 ± 4.398	1.717 (1.662- 1.774, p=0.0)	0.5 405	0.0 17	32. 72 4	0. 00 0	0.5 08	0.5 73	-	-
fibrinoge n	3.602 ± 0.464	3.518 ± 0.663	0.675 (0.585- 0.778, p=0.0)	- 0.3 931	0.0 73	- 5.4 08	0. 00 0	- 0.5 36	- 0.2 51	-	-
ck	195.00 7 ± 273.21 2	1029.5 94 ± 872.8	1.002 (1.002- 1.002, p=0.0)	0.0 024	6.1 5e- 05	38. 32 6	0. 00 0	0.0 02	0.0 02	-	-
ldh	215.35 7 ± 75.036	296.45 5 ± 111.28 2	1.005 (1.005- 1.006, p=0.0)	0.0 053	0.0 00	21. 42 4	0. 00 0	0.0 05	0.0 06	-	-
hbdh	175.90 6 ± 48.18	209.29 5 ± 57.826	1.006 (1.005- 1.007, p=0.0)	0.0 062	0.0 00	15. 63 7	0. 00 0	0.0 05	0.0 07	-	-
ima	75.458 ± 12.891	81.624 ± 8.559	1.015 (1.012- 1.017, p=0.0)	0.0 147	0.0 01	10. 70 7	0. 00 0	0.0 12	0.0 17	-	-
lactate	2.505 ± 0.411	2.825 ± 0.376	3.12 (2.784- 3.494, p=0.0)	1.1 377	0.0 58	19. 58 7	0. 00 0	1.0 24	1.2 51	-	-
anion_ga p	12.345 ± 1.368	13.026 ± 1.456	1.344 (1.29- 1.399, p=0.0)	0.2 953	0.0 21	14. 36 8	0. 00 0	0.2 55	0.3 36	-	-
tco2	22.781	22.739	0.972 (0.921-	- 0.0	0.0	- 1.0	0. 0.	- 0.0	- 0.0	-	-

Table 2. (continued)

	± 1.225	± 1.025	1.025, p=0.293)	287	27	51	29 3	82	25		
			1.342 (1.318- 1.368, p=0.0)	0.2 942	0.0 10	30. 95 7	0. 00 0	0.2 76	0.3 13	-	-
nihss	7.886 ± 2.871	11.529 ± 2.564									
uremia_ 0	20618 (99.177 %)	934 (97.904 %)	-	-	-	-	-	-	-	4.334% (934 / 21552)	95.666% (20618 / 21552)
uremia_ 1	171 (0.823 %)	20 (2.096 %)	2.582 (1.618- 4.121, p=0.0)	0.9 485	0.2 39	3.9 74	0. 00 0	0.4 81	1.4 16	10.471% (20 / 191)	89.529% (171 / 191)
dvt_0	19534 (93.963 %)	847 (88.784 %)	-	-	-	-	-	-	-	4.156% (847 / 20381)	95.844% (19534 / 20381)
dvt_1	1255 (6.037 %)	107 (11.216 %)	1.966 (1.595- 2.423, p=0.0)	0.6 761	0.1 07	6.3 40	0. 00 0	0.4 67	0.8 85	7.856% (107 / 1362)	92.144% (1255 / 1362)
fatty_live r_0	16655 (80.114 %)	812 (85.115 %)	-	-	-	-	-	-	-	4.649% (812 / 17467)	95.351% (16655 / 17467)
fatty_live r_1	4134 (19.886 %)	142 (14.885 %)	0.705 (0.587- 0.845, p=0.0)	- 0.3 502	0.0 93	- 3.7 82	0. 00 0	- 0.5 32	- 0.1 69	3.321% (142 / 4276)	96.679% (4134 / 4276)
diabetes _0	13737 (66.078 %)	617 (64.675 %)	-	-	-	-	-	-	-	4.298% (617 / 14354)	95.702% (13737 / 14354)
diabetes _1	7052 (33.922 %)	337 (35.325 %)	1.064 (0.929- 1.219, p=0.371)	0.0 620	0.0 69	0.8 95	0. 37 1	- 0.0 74	0.1 98	4.561% (337 / 7389)	95.439% (7052 / 7389)
hyperten sion_0	6497 (31.252 %)	290 (30.398 %)	-	-	-	-	-	-	-	4.273% (290 / 6787)	95.727% (6497 / 6787)
hyperten sion_1	14292 (68.748 %)	664 (69.602 %)	1.041 (0.904- 1.198, p=0.578)	0.0 400	0.0 72	0.5 56	0. 57 8	- 0.1 01	0.1 81	4.44% (664 / 14956)	95.56% (14292 / 14956)
coronary _disease _0	11288 (54.298 %)	599 (62.788 %)	-	-	-	-	-	-	-	5.039% (599 / 11887)	94.961% (11288 / 11887)

Table 2. (continued)

coronary_disease_1	9501 (45.702 %)	355 (37.212 %)	0.704 (0.616-0.805, p=0.0)	- 508	0.3 68	- 5.1 28	0. 00	- 0.4 85	- 0.2 17	3.602% (355 / 9856)	96.398% (9501 / 9856)
atrial_fibrillation_0	18811 (90.485 %)	838 (87.841 %)	-	-	-	-	-	-	-	4.265% (838 / 19649)	95.735% (18811 / 19649)
atrial_fibrillation_1	1978 (9.515 %)	116 (12.159 %)	1.316 (1.078-1.608, p=0.007)	0.2 749	0.1 02	2.6 99	0. 00	0.0 75	0.4 75	5.54% (116 / 2094)	94.46% (1978 / 2094)
hyperuricemia_0	18547 (89.215 %)	801 (83.962 %)	-	-	-	-	-	-	-	4.14% (801 / 19348)	95.86% (18547 / 19348)
hyperuricemia_1	2242 (10.785 %)	153 (16.038 %)	1.58 (1.322-1.889, p=0.0)	0.4 575	0.0 91	5.0 27	0. 00	0.2 79	0.6 36	6.388% (153 / 2395)	93.612% (2242 / 2395)
hyperlipidaemia_0	16439 (79.075 %)	801 (83.962 %)	-	-	-	-	-	-	-	4.646% (801 / 17240)	95.354% (16439 / 17240)
hyperlipidaemia_1	4350 (20.925 %)	153 (16.038 %)	0.722 (0.605-0.861, p=0.0)	- 0.3 259	0.0 90	- 3.6 27	0. 00	- 0.5 02	- 0.1 50	3.398% (153 / 4503)	96.602% (4350 / 4503)
hypoproteinemia_0	18479 (88.888 %)	774 (81.132 %)	-	-	-	-	-	-	-	4.02% (774 / 19253)	95.98% (18479 / 19253)
hypoproteinemia_1	2310 (11.112 %)	180 (18.868 %)	1.86 (1.573-2.201, p=0.0)	0.6 208	0.0 86	7.2 48	0. 00	0.4 53	0.7 89	7.229% (180 / 2490)	92.771% (2310 / 2490)
cerebral_hernia_0	20626 (99.216 %)	934 (97.904 %)	-	-	-	-	-	-	-	4.332% (934 / 21560)	95.668% (20626 / 21560)
cerebral_hernia_1	163 (0.784 %)	20 (2.096 %)	2.71 (1.696-4.332, p=0.0)	0.9 968	0.2 39	4.1 66	0. 00	0.5 28	1.4 66	10.929% (20 / 183)	89.071% (163 / 183)
hydrocephalus_0	20565 (98.923 %)	895 (93.816 %)	-	-	-	-	-	-	-	4.171% (895 / 21460)	95.829% (20565 / 21460)
hydrocephalus_1	224 (1.077 %)	59 (6.184 %)	6.052 (4.509-8.125,	1.8 004	0.1 50	11. 98	0. 00	1.5 06	2.0 95	20.848% (59 / 283)	79.152% (224 / 283)

Table 2. (continued)

			p=0.0)				2	0				
frontal_lo be_0	19943 (95.931 %)	868 (90.985 %)	-	-	-	-	-	-	-	-	4.171% (868 / 20811)	95.829% (19943 / 20811)
frontal_lo be_1	846 (4.069 %)	86 (9.015 %)	2.336 (1.852- 2.945, p=0.0)	0.8 483	0.1 18	7.1 66	0. 00	0.6 16	1.0 80		9.227% (86 / 932)	90.773% (846 / 932)
parietal_l obe_0	20180 (97.071 %)	884 (92.662 %)	-	-	-	-	-	-	-		4.197% (884 / 21064)	95.803% (20180 / 21064)
parietal_l obe_1	609 (2.929 %)	70 (7.338 %)	2.624 (2.03- 3.391, p=0.0)	0.9 647	0.1 31	7.3 75	0. 00	0.7 08	1.2 21		10.309% (70 / 679)	89.691% (609 / 679)
temporal _lobe_0	20209 (97.21 %)	886 (92.872 %)	-	-	-	-	-	-	-		4.2% (886 / 21095)	95.8% (20209 / 21095)
temporal _lobe_1	580 (2.79%)	68 (7.128 %)	2.674 (2.063- 3.469, p=0.0)	0.9 836	0.1 33	7.4 13	0. 00	0.7 24	1.2 44		10.494% (68 / 648)	89.506% (580 / 648)
occipital _lobe_0	20422 (98.235 %)	919 (96.331 %)	-	-	-	-	-	-	-		4.306% (919 / 21341)	95.694% (20422 / 21341)
occipital _lobe_1	367 (1.765 %)	35 (3.669 %)	2.119 (1.489- 3.016, p=0.0)	0.7 511	0.1 80	4.1 70	0. 00	0.3 98	1.1 04		8.706% (35 / 402)	91.294% (367 / 402)
insular_l obe_0	20519 (98.701 %)	938 (98.323 %)	-	-	-	-	-	-	-		4.372% (938 / 21457)	95.628% (20519 / 21457)
insular_l obe_1	270 (1.299 %)	16 (1.677 %)	1.296 (0.78- 2.155, p=0.317)	0.2 595	0.2 59	1.0 00	0. 31	- 0.2	0.7 49		5.594% (16 / 286)	94.406% (270 / 286)
range_lo be_0	19559 (94.083 %)	830 (87.002 %)	-	-	-	-	-	-	-		4.071% (830 / 20389)	95.929% (19559 / 20389)
range_lo be_1	467 (2.246 %)	43 (4.507 %)	2.17 (1.576- 2.989, p=0.0)	0.7 746	0.1 63	4.7 45	0. 00	0.4 55	1.0 95		8.431% (43 / 510)	91.569% (467 / 510)
range_lo	329 (1.583)	32 (3.354)	2.292 (1.584-	0.8	0.1	4.3	0.	0.4	1.1		8.864% (32 /	91.136% (329 /

Table 2. (continued)

be_2	%)	%)	3.317, p=0.0)	294	89	99	00 0	60	99	361)	361)
range_lo be_3	224 (1.077 %)	31 (3.249 %)	3.261 (2.226- 4.778, p=0.0)	1.1 821	0.1 95	6.0 66	0. 00 0	0.8 00	1.5 64	12.157% (31 / 255)	87.843% (224 / 255)
range_lo be_4	175 (0.842 %)	15 (1.572 %)	2.02 (1.186- 3.438, p=0.01)	0.7 030	0.2 71	2.5 91	0. 01 0	0.1 71	1.2 35	7.895% (15 / 190)	92.105% (175 / 190)
range_lo be_5	35 (0.168 %)	3 (0.314 %)	2.02 (0.62- 6.58, p=0.243)	0.7 030	0.6 03	1.1 67	0. 24 3	- 0.4 78	1.8 84	7.895% (3 / 38)	92.105% (35 / 38)
basal_ga nglia_0	19869 (95.575 %)	893 (93.606 %)	-	-	-	-	-	-	-	4.301% (893 / 20762)	95.699% (19869 / 20762)
basal_ga nglia_1	920 (4.425 %)	61 (6.394 %)	1.475 (1.129- 1.927, p=0.004)	0.3 888	0.1 37	2.8 47	0. 00 4	0.1 21	0.6 56	6.218% (61 / 981)	93.782% (920 / 981)
brainste m_0	20532 (98.764 %)	938 (98.323 %)	-	-	-	-	-	-	-	4.369% (938 / 21470)	95.631% (20532 / 21470)
brainste m_1	257 (1.236 %)	16 (1.677 %)	1.363 (0.819- 2.268, p=0.234)	0.3 095	0.2 60	1.1 91	0. 23 4	- 0.2 00	0.8 19	5.861% (16 / 273)	94.139% (257 / 273)
epencep halon_0	20362 (97.946 %)	934 (97.904 %)	-	-	-	-	-	-	-	4.386% (934 / 21296)	95.614% (20362 / 21296)
epencep halon_1	427 (2.054 %)	20 (2.096 %)	1.021 (0.649- 1.606, p=0.928)	0.0 209	0.2 31	0.0 90	0. 92 8	- 0.4 32	0.4 74	4.474% (20 / 447)	95.526% (427 / 447)
paravent ricular_0	19786 (95.175 %)	899 (94.235 %)	-	-	-	-	-	-	-	4.346% (899 / 20685)	95.654% (19786 / 20685)
paravent ricular_1	1003 (4.825 %)	55 (5.765 %)	1.207 (0.912- 1.597, p=0.187)	0.1 880	0.1 43	1.3 18	0. 18 7	- 0.0 92	0.4 68	5.198% (55 / 1058)	94.802% (1003 / 1058)
centrum _semiov ale_0	20207 (97.2%)	922 (96.646 %)	-	-	-	-	-	-	-	4.364% (922 / 21129)	95.636% (20207 / 21129)

Table 2. (continued)

centrum _semiov ale_1	582 (2.8%)	32 (3.354 %)	1.205 (0.839-1.73, p=0.313)	0.1 865	0.1 85	1.0 10	0. 31 3	- 0.1 75	0.5 48	5.212% (32 / 614)	94.788% (582 / 614)
thalamus _0	20565 (98.923 %)	937 (98.218 %)	-	-	-	-	-	-	-	4.358% (937 / 21502)	95.642% (20565 / 21502)
thalamus _1	224 (1.077 %)	17 (1.782 %)	1.666 (1.013-2.74, p=0.044)	0.5 102	0.2 54	2.0 11	0. 04 4	0.0 13	1.0 08	7.054% (17 / 241)	92.946% (224 / 241)
aca_0	20524 (98.725 %)	941 (98.637 %)	-	-	-	-	-	-	-	4.384% (941 / 21465)	95.616% (20524 / 21465)
aca_1	265 (1.275 %)	13 (1.363 %)	1.07 (0.611- 1.874, p=0.813)	0.0 676	0.2 86	0.2 36	0. 81 3	- 0.4 93	0.6 28	4.676% (13 / 278)	95.324% (265 / 278)
mca_0	19998 (96.195 %)	912 (95.597 %)	-	-	-	-	-	-	-	4.362% (912 / 20910)	95.638% (19998 / 20910)
mca_1	791 (3.805 %)	42 (4.403 %)	1.164 (0.848- 1.598, p=0.348)	0.1 521	0.1 62	0.9 39	0. 34 8	- 0.1 65	0.4 69	5.042% (42 / 833)	94.958% (791 / 833)
pca_0	20729 (99.711 %)	952 (99.79 %)	-	-	-	-	-	-	-	4.391% (952 / 21681)	95.609% (20729 / 21681)
pca_1	60 (0.289 %)	2 (0.21%)	0.726 (0.177- 2.974, p=0.656)	- 0.3 205	0.7 20	- 0.4 45	0. 65 6	- 1.7 31	1.0 90	3.226% (2 / 62)	96.774% (60 / 62)
va_0	20159 (96.97 %)	927 (97.17 %)	-	-	-	-	-	-	-	4.396% (927 / 21086)	95.604% (20159 / 21086)
va_1	630 (3.03%)	27 (2.83%)	0.932 (0.631- 1.377, p=0.724)	- 0.0 704	0.1 99	- 0.3 53	0. 72 4	- 0.4 61	0.3 20	4.11% (27 / 657)	95.89% (630 / 657)
ba_0	20605 (99.115 %)	945 (99.057 %)	-	-	-	-	-	-	-	4.385% (945 / 21550)	95.615% (20605 / 21550)
ba_1	184 (0.885 %)	9 (0.943 %)	1.067 (0.544-2.09, p=0.851)	0.0 644	0.3 43	0.1 88	- 0. 85	- 0.6 08	0.7 37	4.663% (9 / 193)	95.337% (184 / 193)

Table 2. (continued)

1											
gender_0	10407 (50.06%)	372 (38.994%)	-	-	-	-	-	-	-	3.451% (372 / 10779)	96.549% (10407 / 10779)
gender_1	10382 (49.94%)	582 (61.006%)	1.568 (1.373-1.791, p=0.0)	0.4500	0.068	6.635	0.00	0.317	0.583	5.308% (582 / 10964)	94.692% (10382 / 10964)
cca_plaque_0	16100 (77.445%)	751 (78.721%)	-	-	-	-	-	-	-	4.457% (751 / 16851)	95.543% (16100 / 16851)
cca_plaque_1	4689 (22.555%)	203 (21.279%)	0.928 (0.792-1.088, p=0.356)	-0.0746	0.081	-0.923	0.035	-0.233	0.084	4.15% (203 / 4892)	95.85% (4689 / 4892)
ica_plaque_0	19392 (93.28%)	878 (92.034%)	-	-	-	-	-	-	-	4.332% (878 / 20270)	95.668% (19392 / 20270)
ica_plaque_1	1397 (6.72%)	76 (7.966%)	1.202 (0.945-1.528, p=0.135)	0.1836	0.123	1.496	0.013	-0.057	0.424	5.16% (76 / 1473)	94.84% (1397 / 1473)
eca_plaque_0	20591 (99.048%)	942 (98.742%)	-	-	-	-	-	-	-	4.375% (942 / 21533)	95.625% (20591 / 21533)
eca_plaque_1	198 (0.952%)	12 (1.258%)	1.325 (0.737-2.382, p=0.347)	0.2812	0.299	0.940	0.034	-0.305	0.868	5.714% (12 / 210)	94.286% (198 / 210)
subcortex_lobe_0	18454 (88.768%)	814 (85.325%)	-	-	-	-	-	-	-	4.225% (814 / 19268)	95.775% (18454 / 19268)
subcortex_lobe_1	2335 (11.232%)	140 (14.675%)	1.359 (1.131-1.634, p=0.001)	0.3070	0.094	3.262	0.001	0.123	0.491	5.657% (140 / 2475)	94.343% (2335 / 2475)

Table 2. (continued)

Feature	0 (N=20789)	1 (N=954)	OR (multivariable)	Coef	Std.Er	z	P> z	[0.025	0.975]
tg	1.536 ± 0.433	1.662 ± 0.484	2.458 (2.069-2.92, p=0.0)	0.899	0.088	10.23	0	0.727	1.071
rbc	4.304 ± 0.324	4.408 ± 0.274	4.731 (3.274-6.837, p=0.0)	1.554	0.188	8.275	0	1.186	1.922
age	66.806 ± 12.597	65.335 ± 13.909	1.012 (1.004-1.021, p=0.003)	0.012	0.004	2.971	0.003	0.004	0.021
ast	26.05 ± 12.823	38.25 ± 18.205	1.048 (1.04-1.055, p=0.0)	0.046	0.004	12.413	0	0.039	0.054
plt	190.132 ± 26.424	180.251 ± 36.939	0.977 (0.973-0.98, p=0.0)	-0.024	0.002	13.375	0	-0.027	-0.02
alt	24.193 ± 10.108	26.827 ± 10.349	0.953 (0.942-0.964, p=0.0)	-0.048	0.006	8.177	0	-0.059	-0.036
ima	75.458 ± 12.891	81.624 ± 8.559	1.006 (1.001-1.012, p=0.014)	0.006	0.003	2.453	0.014	0.001	0.012
ldh	215.357 ± 75.036	296.455 ± 111.282	0.984 (0.982-0.987, p=0.0)	-0.016	0.001	12.992	0	-0.018	-0.014
tt	16.432 ± 0.615	16.636 ± 0.809	1.13 (1.009-1.265, p=0.034)	0.122	0.058	2.116	0.034	0.009	0.235
crp	15.314 ± 18.865	55.681 ± 48.823	1.032 (1.028-1.036, p=0.0)	0.031	0.002	15.585	0	0.027	0.035
wbc	8.316 ± 1.286	11.79 ± 3.084	2.091 (1.985-2.204, p=0.0)	0.738	0.027	27.583	0	0.685	0.79

Table 3.

Multivariable logistic regression results

ck	195.007 ± 273.212	1029.59 4 ± 872.8	1.001 (1.001- 1.001, p=0.0)	0.00 1	0	7.86	0	0.001	0.00 1
subcortex_lobe_ 0	18454 (88.768 %)	814 (85.325 %)	-	-	-	-	-	-	-
subcortex_lobe_ 1	2335 (11.232 %)	140 (14.675 %)	1.188 (0.827- 1.707 ,p=0.35 2)	0.17 2	0.185	0.93	0.35 2	- 0.191	0.53 5
frontal_lobe_0	19943 (95.931 %)	868 (90.985 %)	-	-	-	-	-	-	-
frontal_lobe_1	846 (4.069%)	86 (9.015%)	4.577 (1.381- 15.17 ,p=0.01 3)	1.52 1	0.611	2.488	0.01 3	0.323	2.71 9
cerebral_hernia_ 0	20626 (99.216 %)	934 (97.904 %)	-	-	-	-	-	-	-
cerebral_hernia_ 1	163 (0.784%)	20 (2.096%)	0.846 (0.387- 1.85 ,p=0.676)	- 0.16 7	0.399	- 0.418	0.67 6	- 0.949	0.61 5
thalamus_0	20565 (98.923 %)	937 (98.218 %)	-	-	-	-	-	-	-
thalamus_1	224 (1.077%)	17 (1.782%)	0.669 (0.327- 1.373 ,p=0.27 3)	- 0.40 1	0.366	- 1.095	0.27 3	- 1.119	0.31 7
occipital_lobe_0	20422 (98.235 %)	919 (96.331 %)	-	-	-	-	-	-	-

Table 3. (continued)

occipital_lobe_1	367 (1.765%)	35 (3.669%)	2.172 (0.741- 6.368 ,p=0.15 7)	0.77 6	0.549	1.414	0.15 7	-0.3	1.85 1
coronary_diseas e_0	11288 (54.298 %)	599 (62.788 %)	-	-	-	-	-	-	-
coronary_diseas e_1	9501 (45.702 %)	355 (37.212 %)	1.408 (1.151- 1.724 ,p=0.00 1)	0.34 2	0.103	3.322	0.00 1	0.14	0.54 5
hypoproteinemia _0	18479 (88.888 %)	774 (81.132 %)	-	-	-	-	-	-	-
hypoproteinemia _1	2310 (11.112 %)	180 (18.868 %)	1.183 (0.9- 1.554 ,p=0.22 8)	0.16 8	0.139	1.206	0.22 8	- 0.105	0.44 1
parietal_lobe_0	20180 (97.071 %)	884 (92.662 %)	-	-	-	-	-	-	-
parietal_lobe_1	609 (2.929%)	70 (7.338%)	6.939 (2.253- 21.375 ,p=0.0 01)	1.93 7	0.574	3.375	0.00 1	0.812	3.06 2
hyperuricemia_0	18547 (89.215 %)	801 (83.962 %)	-	-	-	-	-	-	-
hyperuricemia_1	2242 (10.785 %)	153 (16.038 %)	0.938 (0.691- 1.275 ,p=0.68 4)	- 0.06 4	0.156	- 0.407	0.68 4	-0.37	0.24 3
temporal_lobe_0	20209 (97.21%)	886 (92.872 %)	-	-	-	-	-	-	-

Table 3. (continued)

temporal_lobe_1	580 (2.79%)	68 (7.128%)	5.242 (1.548- 17.752 ,p=0.008)	1.65 7	0.622	2.662	0.00 8	0.437	2.87 6
range_lobe_0	19559 (94.083%)	830 (87.002%)	-	-	-	-	-	-	-
range_lobe_1	467 (2.246%)	43 (4.507%)	0.359 (0.111- 1.159 ,p=0.087)	- 1.02 5	0.598	- 1.713	0.08 7	- 2.197	0.14 7
range_lobe_2	329 (1.583%)	32 (3.354%)	0.084 (0.01- 0.703 ,p=0.022)	- 2.47 9	1.085	- 2.285	0.02 2	- 4.605	- 0.35 2
range_lobe_3	224 (1.077%)	31 (3.249%)	0.011 (0.0- 0.231 ,p=0.004)	- 4.54 2	1.569	- 2.895	0.00 4	- 7.617	- 1.46 7
range_lobe_4	175 (0.842%)	15 (1.572%)	0.001 (0.0- 0.057 ,p=0.001)	- 6.66 6	1.943	-3.43	0.00 1	- 10.47 5	- 2.85 7
range_lobe_5	35 (0.168%)	3 (0.314%)	0.001 (0.0- 0.115 ,p=0.004)	- 6.58 6	2.259	- 2.915	0.00 4	- 11.01 4	- 2.15 9
hydrocephalus_0	20565 (98.923%)	895 (93.816%)	-	-	-	-	-	-	-
hydrocephalus_1	224 (1.077%)	59 (6.184%)	3.251 (1.939- 5.451 ,p=0.0)	1.17 9	0.264	4.471	0	0.662	1.69 6
gender_0	10407 (50.06%)	372 (38.994%)	-	-	-	-	-	-	-

Table 3. (continued)

gender_1	10382 (49.94%)	582 (61.006)	0.572 (0.454- 0.72 ,p=0.0)	- 0.55 8	0.117	- 4.753	0	- 0.789	- 0.32 8
uremia_0	20618 (99.177)	934 (97.904)	-	-	-	-	-	-	-
uremia_1	171 (0.823%)	20 (2.096%)	1.979 (1.098- 3.564 ,p=0.02 3)	0.68 2	0.3	2.273	0.02 3	0.094	1.27 1
atrial_fibrillation_0	18811 (90.485)	838 (87.841)	-	-	-	-	-	-	-
atrial_fibrillation_1	1978 (9.515%)	116 (12.159)	1.446 (1.087- 1.923 ,p=0.01 1)	0.36 9	0.145	2.534	0.01 1	0.084	0.65 4
basal_ganglia_0	19869 (95.575)	893 (93.606)	-	-	-	-	-	-	-
basal_ganglia_1	920 (4.425%)	61 (6.394%)	1.024 (0.642- 1.633 ,p=0.92 1)	0.02 4	0.238	0.099	0.92 1	- 0.443	0.49
dvt_0	19534 (93.963)	847 (88.784)	-	-	-	-	-	-	-
dvt_1	1255 (6.037%)	107 (11.216)	1.254 (0.922- 1.706 ,p=0.14 9)	0.22 7	0.157	1.443	0.14 9	- 0.081	0.53 4
hyperlipidaemia_0	16439 (79.075)	801 (83.962)	-	-	-	-	-	-	-

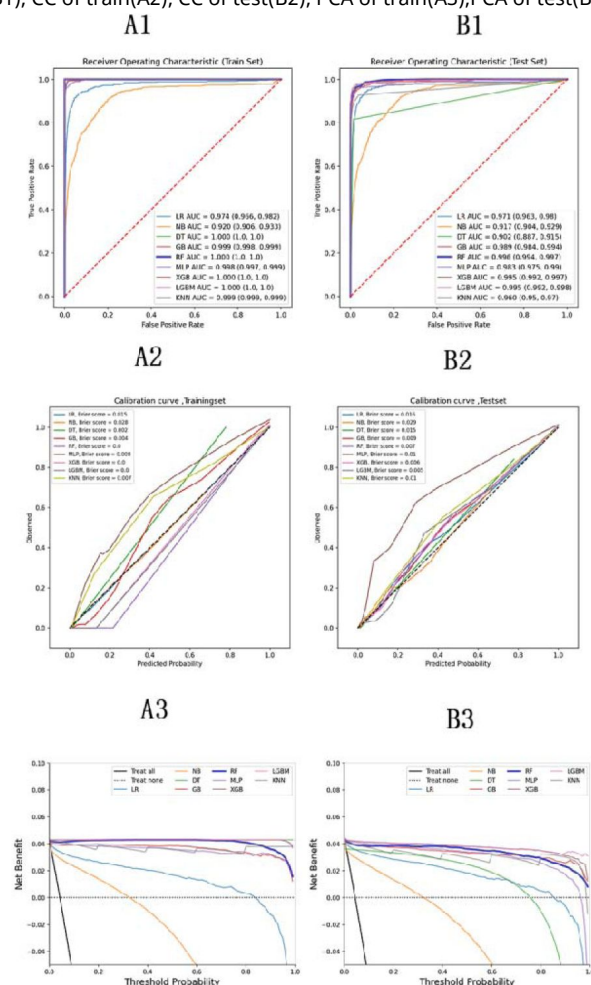
Table 3. (continued)

Table 3. (continued)

hyperlipidaemia_1	4350 (20.925 %)	153 (16.038 %)	0.825 (0.646- 1.052 ,p=0.12 1)	- 0.19 3	- 0.124 1.552	- 0.12 1	- - 0.437	0.05 - 1
fatty_liver_0	16655 (80.114 %)	812 (85.115 %)	-	-	-	-	-	-
fatty_liver_1	4134 (19.886 %)	142 (14.885 %)	0.759 (0.59- 0.978 ,p=0.03 3)	- 0.27 5	- 0.129 2.135	- 0.03 3	- - 0.528	- 0.02 3

Figure 3.

ROC of train(A1), ROC of test(B1), CC of train(A2), CC of test(B2), PCA of train(A3),PCA of test(B3)



Model	AUC	Accuracy	Sensitivity/Recall	Specificity	F1-score	PPV/precision	NPV
Logistic Regression	0.961 0.967	0.968 0.969	0.797 0.788	0.976 0.977	0.687 0.683	0.603 0.603	0.990 0.990
Naive Bayes	0.922 0.918	0.926 0.925	0.672 0.652	0.937 0.937	0.447 0.425	0.334 0.316	0.984 0.984
Decision Tree Classifier	1.000 0.915	1.000 0.978	1.000 0.845	1.000 0.984	1.000 0.770	1.000 0.707	1.000 0.993
GradientBoosting	0.999 0.992	0.995 0.991	0.931 0.875	0.998 0.996	0.943 0.893	0.955 0.911	0.997 0.994
Random Forest Classifier	1.000 0.997	1.000 0.993	1.000 0.853	1.000 0.999	1.000 0.911	1.000 0.977	1.000 0.994
MLP	0.995	0.994	0.939 0.908	0.996	0.933	0.926 0.874	0.997

Table 4.

The AUC, Accuracy, Sensitivity, Specificity, F1-score, positive predictive value (PPV) and negative predictive value (NPV) of ML models of train|test groups

	0.98 5	0.990		0.994	0.891		0.99 6
XGBoost	1.00 0 0.99 7	1.000 0.995	1.000 0.906	1.000 0.999	1.000 0.935	1.000 0.965	1.00 0 0.99 6
LightGBM	1.00 0 0.99 6	1.000 0.995	1.000 0.904	1.000 0.999	1.000 0.933	1.000 0.965	1.00 0 0.99 6
KNeighbors	0.99 9 0.94 9	0.989 0.982	0.960 0.851	0.990 0.987	0.886 0.797	0.823 0.750	0.99 8 0.99 3

Table 4. (continued)

of the first person to show the influence of different features of the first person, we can see that long APTT time contribute best to PSE, then the AST level and others, the NIHSS score may be low and contribute opposite to the final result. Then the decision plot is a collection of model decisions that show how complex models arrive at their predictions.

Discussion

Our study utilized comprehensive clinical data, imaging data, laboratory test data, from the database of the stroke patients and employed machine learning algorithms to establish a predictive model, achieving an AUC score of above 0.95, which demonstrated more accurate predictions compared to traditional statistical methods. Our research found that tree-based ensemble models showed superior overall prediction capabilities when dealing with large sample sizes and high-dimensional features.

During the modeling process, due to the extreme imbalance between negative and positive samples, we employed SMOTEENN technique to resample an imbalanced dataset for machine learning, resulting in improved training performance. Through SHAP analysis, we conducted interpretability analysis of the model and determined the importance of different features.

In our study, age and NIHSS score were treated as continuous variables. We found that, overall, female patients, older patients, and those with higher NIHSS scores were more prone to develop PSE, which is consistent with recent articles. High NIHSS scores, indicative of more severe stroke, increased the risk of complications, ranking only to white blood cell count and d-dimer in our model [5][19][10][20]. However, there are conflicting opinions regarding the impact of age. Some researchs [5][21] suggested that age <65 is a high-risk factor, which aligns with our findings, while some studies [22] confirmed that advanced age is the determining factor. Yamada et al. [21] also concurred with our study in identifying a higher risk of complications among females, whereas Waafi et al. [10] indicated that the likelihood of male patients developing complications is 3.325 times that of females, which contradicts our findings.

Previous studies have shown that patients with diabetes, dyslipidemia, hypertension, depression, or dementia are at an increased risk of developing vascular epilepsy [12]. In our study, statistics and multiple ML models analyzed the association between comorbidities and complications, revealing that patients with coronary heart disease, diabetes, fatty liver, hyperlipidemia, or large artery stenosis or plaques(CCA and ICA) were less likely to develop epilepsy. According to the TOAST classification, ischemic stroke is categorized into five types: large artery atherosclerosis, cardioembolism, small vessel occlusion, other determined etiology, and undetermined etiology. Patients with combined comorbidities generally fall into the categories of large artery atherosclerosis and cardioembolism, which are relatively well-defined and easier to intervene, thus resulting in a lower likelihood of developing epilepsy. Conversely, strokes with undetermined etiology usually have a poor prognosis and are more likely to lead to epilepsy. Among diabetes patients, higher HbA1c levels indicate poorer blood sugar control, resulting in a higher probability of developing complications, which significantly affects certain patients, while those with good control have a lower overall risk of developing complications.

Alain et al. found that cortical infarction was more likely to result in epilepsy in patients hospitalized with anterior circulation ischemic stroke [23]. Lin et al. found that factors such as cortical involvement and intracerebral hemorrhage volume increased the likelihood of PSE, which is consistent with our research findings [8]. Al-Sahli et al. also suggested that cortical brain injury and large-area lesions increased the risk of PSE [5][21]. In our study, statistics showed affections of cortical and subcortical regions both increased the possibility of PSE, but had lower affection than the other features so didn't be selected in lasso regression.

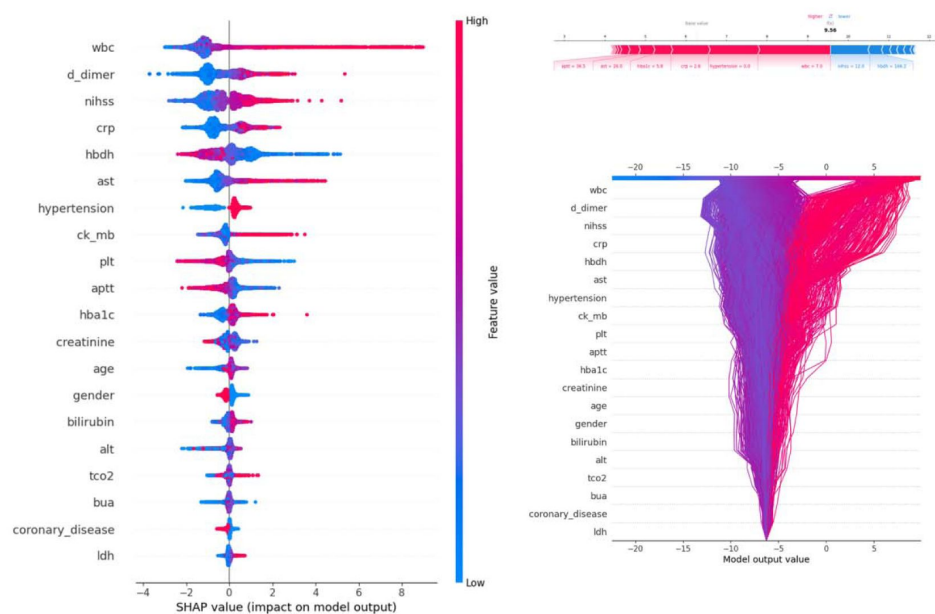


Figure 4.

SHAP value(left),force plot(upper right) and decision plot (lower right)

Previous studies have found that acute infection is a risk factor for ischemic stroke [24]. C-reactive protein (CRP) reflects the level of inflammation and is an independent prognostic factor [25]. In our study, regression and SHAP analysis both showed that white blood cell count had great impact among the routine blood test parameters, in SHAP it even surpassed the NIHSS score. High white blood cell count may indicate severe inflammation and infection, as well as increased blood viscosity, making patients more susceptible to secondary complications. In general, high red blood cell count and low platelet count also have some influence.

A large-scale study on Chinese individuals found a negative correlation between plasma high-density lipoprotein cholesterol (HDL-C) concentration and the risk of ischemic stroke, a weak positive correlation between plasma triglyceride (TG) concentration and the risk of ischemic stroke, and a strong correlation between plasma low-density lipoprotein cholesterol (LDL-C) concentration and apolipoprotein B [26]. High HDL-C levels are associated with better prognosis [27]. Our study is consistent with previous research, indicating that high LDL-C, low HDL-C, and high TG levels are more likely to lead to PSE. This can be easily understood as high cholesterol and triglyceride levels lead to increased blood viscosity and vascular sclerosis, making it easier for clots to form [12][28][29]. Higher D-dimer levels indicate greater brain tissue damage and a higher likelihood of PSE. Overall, lower activated partial thromboplastin time (APTT) and fibrinogen levels are associated with an increased risk of PSE. INR, PT, and TT have a smaller impact. Among liver function parameters, aspartate aminotransferase (AST) has the greatest influence on PSE, while high AST levels, low alanine aminotransferase (ALT) levels, and low albumin levels all have a certain degree of impact. Lingling Ding et al. found that liver enzyme subgroups characterized by alanine aminotransferase and aspartate aminotransferase were associated with a high risk of adverse function [30], which is consistent with our research.

Studies have shown that subgroups identified by renal function biomarkers such as urinary microalbumin, cystatin C, and creatinine have significantly higher stroke recurrence and poorer prognosis [30]. In our study, low urea levels and high uric acid levels had a negative impact [31][32][33]. Our research is similar to their conclusions. While elevated uric acid levels at admission are positively associated with PSE, patients previously diagnosed with hyperuricemia are less likely to develop epilepsy. Considering that uric acid functions as a strong reducing agent and has neuroprotective properties [34], patients with normal liver and kidney function and a certain degree of hyperuricemia have stronger resistance to emergencies [35][36]. However, excessively high uric acid levels indicate metabolic disorders and poor liver and kidney function, which are associated with poor prognosis.

When stroke patients are admitted, cardiac enzyme profile tests are often performed to rule out concurrent myocardial ischemia. However, studies have shown that elevated CK-MB in stroke patients may not only be related to the heart [37]. Multiple cardiac enzymes are important prognostic indicators [38][39] and have been included in stroke scores [40]. Some studies have shown a higher incidence of abnormal serum cardiac enzyme profiles in the acute phase of stroke. Although the incidence of abnormalities is unrelated to the nature of the stroke, it is associated with the severity of the stroke, with patients with consciousness disorders having a significantly higher incidence of abnormal cardiac enzyme profiles than those without consciousness disorders [41]. In our study, CK, CK-MB, and IMA in the cardiac enzyme profile had a significant impact and high predictive value, but the specific mechanisms require further research [34].

Although our study incorporates a large amount of information and utilizes almost all available data, including clinical data, imaging data, and laboratory test data, in an attempt to establish more accurate prediction models beyond traditional statistics using machine learning algorithms, there are still several limitations in the modeling process.

While the current study provides valuable conclusion, the data sample may not be fully representative, and the model's generalization ability requires further assessment. Although the data was collected from multiple tertiary hospitals, encompassing over 20,000 cases, earlier data was lost due to hospital system upgrades. The collected data mainly represents patients diagnosed in the past five years and is primarily concentrated in the Chongqing region, which may limit the model's applicability to other geographic areas.

Additionally, the retrospective nature of the research has resulted in the lack of certain important predictive indicators. As this was a retrospective study, many potentially meaningful features, such as hemorheology, thromboelastography, and hormone levels, were significantly missing and had to be excluded. Incorporating these additional features could potentially improve the model's accuracy.

To enhance the predictive power of the model, it would be beneficial to incorporate more beyond baseline patient characteristics. For example, the current analysis primarily utilized the results of the first examination upon admission, without fully leveraging the information from subsequent examinations. In future research, the use of recurrent neural networks could facilitate the comprehensive extraction of features from the entire sequence of examinations.

To further strengthen the study, data standardization should be improved, and the number of cases and important indicators should continue to increase. Additionally, it would be advisable to explore more advanced scientific methods, such as deep learning, and fully leverage all available data to make more accurate predictions.

Conclusion

We developed an interpretable machine learning model to predict the risk of post-stroke epilepsy (PSE) in hospitalized patients with ischemic stroke. Leveraging a large volume of medical records, our artificial intelligence model demonstrates good predictive performance for PSE. The key predictors identified by the model include NIHSS score, D-dimer levels, lactate levels, and white blood cell count, followed by indicators related to liver function and cardiac enzyme profiles. The transparency and interpretability of the model's predictions can foster trust among clinical practitioners and facilitate decision-making. While the results are promising, further prospective studies are needed to validate the clinical utility of this tool before its application in real-world settings.

Data availability statement

The codes, models, analysis results was uploaded at <https://github.com/conanan/lasso-ml>¹. The full dataset can be provided for researchers if needed by the corresponding author.

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Additional information

Author Contributions

JL and HH are both first writer who analysed the data by python and wrote the first draft of the manuscript. YD and TS are both corresponding author who wrote part of the draft and designed the original research. Chongqing University Central Hospital, Chongqing University Qianjiang Hospital, Yubei District hospital and Bishan hospital of Chongqing Medical University provided the database of all cases of the patients. The others collected data and wrote sections of the manuscript. All authors took part in the research and contributed to manuscript revision, read, and approved the submitted version.

Ethics approval statement

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

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Conflict of interests

The authors have no relevant conflicts of interest to disclose.

Patient consent statement

This study was a retrospective study and only deidentified patient data were collected, exempting the need for patient informed consent rights.

Clinical trial registration

The trail number is RS202406.

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Editors

Reviewing Editor

Alan Talevi

National University of La Plata, La Plata, Argentina

Senior Editor

Eduardo Franco

McGill University, Montreal, Canada

Reviewer #1 (Public review):

Summary:

This is a large cohort of ischemic stroke patients from a single centre. The author successfully set up predictive models for PTS.

Strengths:

The design and implementation of the trial are acceptable, with the credibility of the results. It may provide evidence of seizure prevention in the field of stroke treatment.

Weaknesses:

My concerns are well responded to.

<https://doi.org/10.7554/eLife.98759.2.sa3>

Reviewer #2 (Public review):

Summary

The authors present multiple machine-learning methodologies to predict post-stroke epilepsy (PSE) from admission clinical data.

Strengths

The Statistical Approach section is very well written. The approaches used in this section are very sensible for the data in question.

Typos have now been addressed and improved interpretability has been added to the paper, which is appreciated.

Weaknesses

The authors have clarified that the first features available for each patient have been used. However, they have not shown that these features did not occur before the time of post-stroke epilepsy. Explicit clarification of this should be performed.

The likely impact of the work on the field

If this model works as claimed, it will be useful for predicting PSE. This has some direct clinical utility.

Analysis of features contributing to PSE may provide clinical researchers with ideas for further research on the underlying aetiology of PSE.

<https://doi.org/10.7554/eLife.98759.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors report the performance of a series of machine learning models inferred from a large-scale dataset and externally validated with an independent cohort of patients, to predict the risk of post-stroke epilepsy. Some of the reported models have very good explicative performance, and seem to have very good predictive ability.

Strengths:

The models have been derived from real-world large-scale data.

Performances of the best-performing models seem to be very good according to the external validation results.

Early prediction of risk of post-stroke epilepsy would be of high interest to implement early therapeutic interventions that could improve prognosis.

Code is publicly available. The authors also stated that the datasets used are available on request.

Weaknesses:

The writing of the article may be significantly improved.

Although the external validation is appreciated, cross-validation to check robustness of the models would also be welcome.

External validation results may be biased/overoptimistic, since the authors informed that "The external validation cohort focused more on collecting positive cases 80 to examine the model's ability to identify positive samples", which may result in overoptimistic PPV and Sensitivity estimations. The specificity for the external validation set has not been disclosed.

<https://doi.org/10.7554/eLife.98759.2.sa1>

Author response:

The following is the authors' response to the current reviews.

Public Reviews:

Reviewer #2 (Public review):

Weaknesses:

The authors have clarified that the first features available for each patient have been used. However, they have not shown that these features did not occur before the time of post-stroke epilepsy. Explicit clarification of this should be performed.

The data utilized in our analysis were collected during the first examination or test conducted after the patients' admission. We specifically excluded any patients with a history of epilepsy, ensuring that all cases of epilepsy identified in our study occurred after admission. Therefore, the features we analyzed were collected after the patients' admission but prior to the onset of post-stroke epilepsy.

Reviewer #3 (Public review):

Weaknesses:

The writing of the article may be significantly improved.

Although the external validation is appreciated, cross-validation to check robustness of the models would also be welcome.

Thank you for your helpful advice. Performing n-fold cross-validation is a crucial step to ensure the reliability and robustness of the reported results, especially when dealing with the datasets which don't have sufficient quantity. We revised our code and did a 5 fold cross-validation version ,it didn't have much promote (because our model has reach the auc of

0.99) .Considering that we have sufficient quantity of more than 20000 records, we think split the dataset by 7:3 and train the model is enough for us. We have uploaded the code of 5 fold cross-validation version and plotted the 5 fold test roc on GitHub at https://github.com/conanan/lasso-ml/lasso_ml_cross_validation.ipynb as an external resource. We trained the 5 fold average model and plotted the 5 fold test roc curves, the results show some improvement, but it is not substantial because the best model are still tree models in the end.

External validation results may be biased/overoptimistic, since the authors informed that "The external validation cohort focused more on collecting positive cases 80 to examine the model's ability to identify positive samples", which may result in overoptimistic PPV and Sensitivity estimations. The specificity for the external validation set has not been disclosed.

Thank you for your valuable feedback regarding the external validation results. We appreciate your concerns about potential bias and overoptimism in our estimations of positive predictive value (PPV) and sensitivity.

To clarify, we have uploaded the code for external validation on GitHub at <https://github.com/conanan/lasso-ml>. The results indicate that the PPV is 0.95 and the specificity is 0.98.

While we focused on collecting more positive cases due to their lower occurrence rate, this approach allows us to better evaluate the model's ability to predict positive samples, which is crucial in clinical settings. We believe that emphasizing positive cases enhances the model's utility for practical applications(So a little overoptimism is acceptable).

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Weaknesses 1:

The methodology needs further consideration. The Discussion needs extensive rewriting.

Thanks for your advice, we have revised the Discussion

Reviewer #2 (Public Review):

Weaknesses 2:

There are many typos and unclear statements throughout the paper.

There are some issues with SHAP interpretation. SHAP in its default form, does not provide robust statistical guarantees of effect size. There is a claim that "SHAP analysis showed that white blood cell count had the greatest impact among the routine blood test parameters". This is a difficult claim to make.

Thank you for your suggestion that the SHAP analysis is really just a means of interpreting the model. In our research, we compared the SHAP analysis with traditional statistical methods, such as regression analysis. We found the SHAP results to be consistent with the statistical results from the regression for variables like white blood cell count (see Table 1). This alignment leads us to believe the SHAP analysis is providing reliable insights in this context

The Data Collection section is very poorly written, and the methodology is not clear.

Thanks for your advice, we have revised the Data Collection section.

There is no information about hyperparameter selection for models or whether a hyperparameter search was performed. Given this, it is difficult to conclude whether one machine learning model performs better than others on this task.

Thank you for the advices of performing hyperparameter. We used the package of sklearn, xgboost, lightgbm of python 3.10 to construct the model and didn't change the default settings before. It is not proper and may lead to less certain conclusions. Now we carry out grid search to select and optimize hyperparameters and they make the model better. The best model is still RF.

The inclusion and exclusion criteria are unclear - how many patients were excluded and for what reasons?

The procedure of selection is in figure1. Total there are 42079 records from the stroke database, 24733 patients were diagnosed as ischemic stroke or lacunar stroke with new onset. Then we excluded hemorrhage stroke(4565), history of stroke(2154), TIA(3570), unclear cause stroke(561) and records who missed important data(6496). Then we excluded patients whose seizure might be attributed to other potential causes (brain tumor, intracranial vascular malformation, traumatic brain injury, etc)(865). Then we exclude patient who had a seizure history(152) or died in hospital (1444). Then we excluded patients who were lost in follow-up (had no outpatient records and can't contact by phone) or died within 3 months of the stroke incident(813). Finally 21459 cases are involved in this research.

There is no sensitivity analysis of the SMOTE methodology: How many synthetic data points were created, and how does the number of synthetic data points affect classification accuracy?

Thanks for your remind, we have accept these advice and change the SMOTE to SMOTEENN (Synthetic Minority Over-sampling Technique combined with Edited Nearest Neighbors) technique to resample an imbalanced dataset for machine learning. The code is

```
smoteenn = SMOTEENN(samplingstrategy='auto', randomstate=42)
```

the SMOTEENN class comes from the imblearn library. The samplingstrategy='auto' parameter tells the algorithm to automatically determine the appropriate sampling strategy based on the class distribution. The randomstate=42 parameter sets a seed for the random number generator, ensuring reproducibility of the results.

Did the authors achieve their aims? Do the results support their conclusions?

Yes, we have achieve some of the aims of predicting PSE while still leave some problem.

The paper does not clarify the features' temporal origins. If some features were not recorded on admission to the hospital but were recorded after PSE occurred, there would be temporal leakage.

The data used in our analysis is from the first examination or test conducted after the patients' admission, retrieved from a PostgreSQL database. First, we extracted the initial admission date for patients admitted due to stroke. Then, we identified the nearest subsequent examination data for each of those patients.

The sql code like follows:

```
SELECT TO_DATE(condition_start_date, 'DD-MM-YYYY') AS DATE
```

```
FROM diagnosis
```

```
WHERE person_id ={} and (condition_name like '%梗死%' or condition_name like '%梗塞%')
and(condition_name like '%脑%' or condition_name like '%腔隙%'))
```

```
order by DATE limit 1
```

The authors claim that their models can predict PSE. To believe this claim, seeing more information on out-of-distribution generalisation performance would be helpful. There is limited reporting on the external validation cohort relative to the reporting on train and test data.

Thank you for the advice. The external validation is certainly very important, but there have been some difficulties in reaching a perfect solution. We have tried using open-source databases like the MIMIC database, but the data there does not fit our needs as closely as the records from our own hospital. The MIMIC database lacks some of the key features we require, and also lacks the detailed patient follow-up information that is crucial for our analysis. Given these limitations, we have decided to collect newer records from the same hospitals here in Chongqing. We believe this will allow us to build a more comprehensive dataset to support robust external validation. While it may not be a perfect solution, gathering this additional data from our local healthcare system is a pragmatic step forward. Looking ahead, we plan to continue expanding this Chongqing-based dataset and report on the results of the greater external validation in the future. We are committed to overcoming the challenges around data availability to strengthen the validity and generalizability of our research findings.

For greater certainty on all reported results, it would be most appropriate to perform n-fold cross-validation, and report mean scores and confidence intervals across the cross-validation splits

Thank you for your helpful advice. Performing n-fold cross-validation is a crucial step to ensure the reliability and robustness of the reported results, especially when dealing with the datasets which don't have sufficient quantity. While we have sufficient quantity of more than 20000 records, so we think split the dataset by 7:3 and train the model is enough for us. We revised our code and did a 5 fold cross-validation version, it had little promote (because our model has reach the auc of 0.99) , we may use this great technique in our next study if there is not enough cases.

Additional context that might help readers

The authors show force plots and decision plots from SHAP values. These plots are non-trivial to interpret, and the authors should include an explanation of how to interpret them.

Thank you for your helpful advice. It is a great improve for our draft, we have added the explanation that we use the force plot of the first person to show the influence of different features of the first person, we can see that long APTT time contribute best to PSE, then the AST level and others, the NIHSS score may be low and contribute opposite to the final result. Then the decision plot is a collection of model decisions that show how complex models arrive at their predictions

Reviewer #3 (Public Review):

Weaknesses3:

There are issues with the readability of the paper. Many abbreviations are not introduced properly and sometimes are written inconsistently. A lot of relevant references are omitted. The methodological descriptions are extremely brief and, sometimes, incomplete.

Thanks for your advice, we have revised these flaws.

The dataset is not disclosed, and neither is the code (although the code is made available upon request). For the sake of reproducibility, unless any bioethical concerns impede it, it would be good to have these data disclosed.

Thank you for your recommendations. We have made the code available on GitHub at <https://github.com/conanan/lasso-ml>. While the data is private and belongs to the hospital. Access can be requested by contacting the corresponding author to apply from the hospitals and specifying the purpose of inquiry.

Although the external validation is appreciated, cross-validation to check the robustness of the models would also be welcome.

Thank you for your valuable advice. Performing n-fold cross-validation is crucial for ensuring the reliability and robustness of results, especially with limited datasets. However, since we have over 20,000 records, we believe that a 70:30 split for training and testing is sufficient.

We revised our code and implemented 5-fold cross-validation, which provided minimal improvement, as our model has already achieved an AUC of 0.99. We plan to use this technique in future studies if we encounter fewer cases.

Recommendations for the authors:**Reviewer #1 (Recommendations For The Authors):**

My comments include two parts:

(1) Methodology

a-This study was based on multiple clinical indicators to construct a model for predicting the occurrence of PSE. It involved various multi-class indicators such as the affected cortical regions, locations of vascular occlusion, NIHSS scores, etc. Only using the SHAP index to explain the impact of multi-class variables on the dependent variable seems slightly insufficient. It might be worth considering the use of dummy variables to improve the model's accuracy.

Thank you for the detailed feedback on the study methodology. The SHAP analysis is really just a means of interpreting the model, which we compared with the combination of SHAP and traditional statistics, so we think SHAP analysis is reliable in this research. We have used the dummy variables, especially when dealing with the affected cortical regions, locations of vascular occlusion, for example if frontal region is involved the variable is 1. But they have less impact in the machine learning model

b-The study used Lasso regression to select 20 features to build the model. How was the optimal number of 20 features determined?

Lasso regression is a commonly used feature screening method. Since we extract information from the database and try to include as many features as possible, the cross-verification curve of lasso regression includes 78 features best, but it will lead to too complex model. We select 10,15,20,25,30 features for modeling according to the experiment. When 20 features are found, the model parameters are good and relatively concise. Improve the number of features contribute little to the model effect, decrease the number of features influence the concise of model ,for example the auc of the model with 15 features will drop under 0.95. So we finally select 20 features.

c-The study indicated that the incidence rate of PSE in the enrolled patients is 4.3%, showing a highly imbalanced dataset. If singly using the SMOTE method for oversampling, could this lead to overfitting?

Thanks for your remind, singly using the SMOTE method for oversampling is improper. Now we have find this improvement and change the SMOTE to SMOTEENN (Synthetic Minority Over-sampling Technique combined with Edited Nearest Neighbors) technique to resample an imbalanced dataset for machine learning. First, oversampling with SMOTE and then undersampling with ENN to remove possible noise and duplicate samples. The code is

```
smoteenn = SMOTEENN(sampling_strategy='auto', random_state=42)
```

the SMOTEENN class comes from the imblearn library. The sampling_strategy='auto' parameter tells the algorithm to automatically determine the appropriate sampling strategy based on the class distribution. The random_state=42 parameter sets a seed for the random number generator, ensuring reproducibility of the results.

(2) Clinical aspects:

Line 8, history of ischemic stroke, this is misexpression, could be: diagnosis of ischemic stroke.

Line 8, several hospitals, should be more exact; how many?

Line 74 indicates that the data are from a single centre, this should be clarified.

Line 4 data collection: The criteria read unclear; please clarify further.

Thanks for your remind, we have revised the draft and correct these errors.

Line 110, lab parameters: Why is there no blood glucose?

Because many patients' blood sugar fluctuates greatly and is easily affected by drugs or diet, we finally consider HBA1c as a reference index by asking experts which is more stable.

Line 295, The author indicated that data lost; this should be clarified in the results part, and further, the treatment of missing data should be clarified in the method part.

Thanks for your remind, we have revised the draft and correct these errors.

I hope to see a table of the cohort's baseline characters. The discussion needs extensive rewriting; the author seems to be swinging from the stoke outcome and the seizure, sometimes losing the target.

Figure1 is the procedure of the selection of patients. Table1 contains the cohort's baseline characters

For the swinging from the stroke outcome and the seizure, that is because there are few articles on predicting epilepsy directly by relevant indicators, while there are more articles on prognosis. So we can only take epilepsy as an important factor in prognosis and comprehensively discuss it, or we can't find enough articles and discuss them

Reviewer #2 (Recommendations For The Authors):

There are typos and examples of text that are not clear, including:

"About the nihss score, the higher the nihss score, the more likely to be PSE, nihss score has a third effect just below white blood cell count and D-dimer."

"and only 8 people made incorrect predictions, demonstrating a good predictive ability of the model."

"female were prone to PSE"

"Wafji's research"

"One-heat' (should be one-hot)

Thanks for your remind, we have revised the draft and correct these errors.

The Data Collection section is poorly written, and the methodology is not clear. It would be much more appropriate to include a table of all features used and an explanation of what these features involve. It would also be useful to see the mean values of these features to assess whether the feature values are reasonable for the dataset.

Thanks for your remind. All data are from the first examination or test after admission, presented through the postgresql database. First we extract the first date of the patients who was admitted by stroke, then we extract informations from the nearest examination from the admission. We extract by the SQL code by computer instead of others who may extract data by manual so we get as much data as possible other than only get the features which was reported before. The table of all features used and their mean±std is in table1.

The paper does not clarify the features' temporal origins. If some features were not recorded on admission to the hospital but were recorded after PSE occurred, there would be temporal leakage. I would need this clarified before believing the authors achieved their claims of building a predictive model.

All relevant index results were from the first examination after admission, and the mean standard deviation was listed in the statistical analysis section in table1.

The authors claim that their models can predict PSE. To believe this claim, seeing more information on out-of-distribution generalisation performance would be helpful. There is limited reporting on the external validation cohort relative to the reporting on train and test data.

Thank you for the advice, the external validation is very important but there are some difficulties to reach a perfect one. We have tried some of the open source database like the mimic database, but these data don't fit our request because they don't have as much features as our hospital and lack of follow-up of the relevant patients. In the end we collected the newer records in the same hospitals in Chongqing and we will collect more and report a greater external validation in the future.

For greater certainty on all reported results, It would be most appropriate to perform n-fold cross-validation, and report mean scores and confidence intervals across the cross-validation splits.

Thank you for your helpful advice. Performing n-fold cross-validation is a crucial step to ensure the reliability and robustness of the reported results, especially when dealing with the datasets which don't have sufficient quantity. While we have sufficient quantity of more than 20000 records, so we think split the dataset by 7:3 and train the model is enough for us. We revised our code and did a 5 fold cross-validation version ,it had little promote, we will use this great technique in our next study.

The authors show force plots and decision plots from SHAP values. These plots are non-trivial to interpret, and the authors should include an explanation of how to interpret them.

It is a great improve for our draft, we have added the explanation we use the force plot of the first person to show the influence of different features of the first person, we can see that long APTT time contribute best to PSE, then the AST level and others, the NIHSS score may be low and contribute lower to the final result. Then the decision plot is a collection of model decisions that show how complex models arrive at their predictions

Reviewer #3 (Recommendations For The Authors):

Abbreviations should not be defined in the abstract)or only in the abstract).

Please explicit what are the purposes of the study you are referring to in "Currently, most studies utilize clinical data to establish statistical models, survival analysis and cox regression."

Authors affirm: "there is still a relative scarcity of research 49 on PSE prediction, with most studies focusing on the analysis of specific or certain risk factors ." This statement is especially curious since the current study uses risk factors as predictors.

It is not clear to me what the authors mean by "No study has proposed or established a more comprehensive and scientifically accurate prediction model." The authors do not summarize the statistical parameters of previously reported model, or other relevant data to assess coverage or validity (maybe including a Table summarizing such information would be appropriate. In any case, I would try to omit statements that imply, to some extent, discrediting previous studies without sufficient foundation.

"antiepileptic drugs" is an outdated name. Please use "antiseizure medications"

Thanks for your remind, we have revised the draft and correct these errors.

The authors say regarding missing data that they "filled the data of the remaining indicators with missing values of more than 1000 cases by random forest algorithm". Please clarify what you mean by "of more than 1000 cases." Also, provide details on the RF model used to fill in missing data.

Thanks for your remind. "of more than 1000 cases" was a wrong sentence and we have corrected it. Here is the procedure, first we counted the values of all laboratory indicators for the first time after stroke admission(everyone who was admitted because of stroke would perform blood routine , liver and kidney function and so on), excluded indicators with missing values of more than 10%, and filled the data of the remaining indicators with missing values by random forest algorithm using the default parameter. First, we go through all the

features, starting with the one with the least missing (since the least accurate information is needed to fill in the feature with the least missing). When filling in a feature, replace the missing value of the other feature with 0. Each time a regression prediction is completed, the predicted value is placed in the original feature matrix and the next feature is filled in. After going through all the features, the data filling is complete.

Please specify what do you mean by negative group and positive group, Avoid tacit assumptions.

Thanks for your remind, we have revised the draft and correct these errors.

Please provide more details (and references) on the smote oversampling method. Indicate any relevant parameters/hyperparameters.

Thanks for your remind, we have accept these advice and change the SMOTE to SMOTEENN (Synthetic Minority Over-sampling Technique combined with Edited Nearest Neighbors) technique to resample an imbalanced dataset for machine learning. The code is

```
smoteenn = SMOTEENN(sampling_strategy='auto', random_state=42)
```

the SMOTEENN class comes from the imblearn library. The sampling_strategy='auto' parameter tells the algorithm to automatically determine the appropriate sampling strategy based on the class distribution. The random_state=42 parameter sets a seed for the random number generator, ensuring reproducibility of the results.

The methodology is presented in an extremely succinct and non-organic manner (e.g., (Model building) Select the 20 features with the largest absolute value of LASSO." Please try to improve the narrative.

Lasso regression is a commonly used feature screening method. Since we extract information from the database and try to include as many features as possible, the cross-verification curve of lasso regression includes 78 features best, but it will lead to too complex model. We select 10,15,20,25,30 features for modeling according to the experiment. When 20 features are found, the model parameters are good and relatively concise. Improve the number of features contribute little to the model effect, decrease the number of features influence the concise of model ,for example the auc of the model with 15 features will drop under 0.95. So we finally select 20 features.

Many passages of the text need references. For example, those that refer to Levene test, Welch's t-test, Brier score, Youden index, and many others (e.g., NIHSS score). Please revise carefully.

Thanks for your remind, we have revised the draft and correct these errors.

"Statistical details of the clinical characteristics of the patients are provided in the table." Which table? Number?

Thanks for your remind, we have revised the draft and correct these errors, it is in table1.

Many abbreviations are not properly presented and defined in the text, e.g., wbc count, hba1c, crp, tg, ast, alt, bilirubin, bua, aptt, tt, d_dimer, ck. Whereas I can guess the meaning, do not assume everyone will. Avoid assumptions.

ROC is sometimes written "ROC" and others, "roc." The same happens for PPV/ppv, and many other words (SMOTE; NIHSS score, etc.).

Please rephrase "ppv value of random forest is the highest, reaching 0.977, which is more accurate for the identification of positive patients(the most important function of our models)". PPV always refer to positive predictions that are corroborated, so the sentences seem redundant.

Thanks for your remind, we have revised the draft and correct these errors.

What do you mean by "Complex algorithms". Please try to be as explicit as possible. The text looks rather cryptic or vague in many passages.

Thanks for your remind, "Complex algorithms" is corrected by machine learning.

The text needs a thorough English language-focused revision, since the sense of some sentences is really misleading. For instance "only 8 people made incorrect predictions,". I guess the authors try to say that the best algorithm only mispredicted 8 cases since no people are making predictions here. Also, regarding that quote... Are the authors still speaking of the results of the random forest model, which was said to be one of the best performances?

Thanks for your remind, we have revised the draft and correct these errors.

The authors say that they used, as predictors "comprehensive clinical data, imaging data, laboratory test data, and other data from stroke patients". However, the total pool of predictors is not clear to me at this point. Please make it explicit and avoid abbreviations.

Thanks for your remind, we have revised the draft and correct these errors.

Although the authors say that their code is available upon request, I think it would be better to have it published in an appropriate repository.

Thanks for your remind, we showed our code at <https://github.com/conanan/lasso-ml>.

<https://doi.org/10.7554/eLife.98759.2.sa0>